

Homework 19

Please continue reading Section 6.2 of Durrett. Start right before Theorem 6.4 (“The next goal of this section is ...”). Continue through Theorem 6.5. Then read Example 6.13 and Example 6.14. Example 6.14 uses the results of this section to derive the formulas for expected value and variance of a binomially distributed random variable. We have been using these formulas since early in the semester but did not learn why they are true.

Please work the following exercises from Durrett:

11 (textbook answer: mean 6, variance 43—show how to calculate these values using the methods of Chapter 6.)

Suppose X and Y are independent random variables with $E[X] = 1$, $E[Y] = 2$, $\text{Var}(X) = 3$, and $\text{Var}(Y) = 1$. We are to find the expected value and variance of the random variable $3X + 4Y - 5$.

$$\begin{aligned} E[3X + 4Y - 5] &= 3E[X] + 4E[Y] - 5 \\ &= 3 + 4(2) - 5 \\ &= 11 - 5 \\ &= 6. \end{aligned}$$

$\text{Var}(3X + 4Y - 5) = \text{Var}(3X) + \text{Var}(4Y)$ since X and Y are independent and 5 is constant.

From Chapter 1 (or the formula for variance), we know $\text{Var}(3X) = 9\text{Var}(X)$ and $\text{Var}(4Y) = 16\text{Var}(Y)$. Therefore, $\text{Var}(3X + 4Y - 5) = 9(3) + 16(1) = 43$.

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Let X be the random variable representing the number of boys who know the answer and Y be the random variable representing the number of girls who know the answer. Under the independence assumption, we have $X \sim \text{binomial}\left(18, \frac{1}{3}\right)$ and $Y \sim \text{binomial}\left(12, \frac{1}{2}\right)$. Therefore, $E[X] = 6$, $\text{Var}(X) = 18\left(\frac{1}{3}\right)\left(\frac{2}{3}\right) = 4$, $E[Y] = 6$, and $\text{Var}(Y) = 12\left(\frac{1}{2}\right)\left(\frac{1}{2}\right) = 3$. We thus have $E[X + Y] = 12$. Under the independence assumption, X and Y are independent, and so $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) = 4 + 3 = 7$.

13 (textbook answer: mean 2, variance ~~4.82~~ 1.72 (misprinted in textbook); 1 boy and 1 girl. Explain how to get these results.)

(a) Let X be the random variable representing the number of boys who get in, and let Y be the random variable representing the number of girls who get in. Assuming the admission events are independent, X and Y are independent, $X \sim \text{binomial}(12, 0.1)$, and $Y \sim \text{binomial}(4, 0.2)$. We thus have $E[X + Y] = E[X] + E[Y] = 12(0.1) + 4(0.2) = 1.2 + 0.8 = 2$ and $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) = 12(0.1)(0.9) + 4(0.2)(0.8) = 1.72$.

(b) The probability of 2 boys and no girls accepted is $P((X = 2) \cap (Y = 0)) = P(X = 2)P(Y = 0) = C_{12,2}(0.1)^2(0.9)^{10} \cdot C_{4,0}(0.8)^4 \approx 0.09426$. The probability of 1 boy and 1 girl accepted is $P((X = 1) \cap (Y = 1)) = P(X = 1)P(Y = 1) = C_{12,1}(0.1)^1(0.9)^{11} \cdot C_{4,1}(0.2)^1(0.8)^3 \approx 0.1542$. The probability of one boy and one girl is higher (15% versus 9%).

The random variable X takes values $-2, -1, 0, 1, 2$, all with probability $1/5$. We let $Y = X^2$.

- (a) We calculate $\text{Cov}(X, Y)$: $\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$. The random variable $XY = X^3$ takes values $-8, -1, 0, 1, 8$, all with probability $1/5$. We thus have $E[XY] = (-8)\left(\frac{1}{5}\right) + (-1)\left(\frac{1}{5}\right) + (0)\left(\frac{1}{5}\right) + (1)\left(\frac{1}{5}\right) + (8)\left(\frac{1}{5}\right) = 0$. We also have $E[X] = (-2)\left(\frac{1}{5}\right) + (-1)\left(\frac{1}{5}\right) + (0)\left(\frac{1}{5}\right) + (1)\left(\frac{1}{5}\right) + (2)\left(\frac{1}{5}\right) = 0$, and so $\text{Cov}(X, Y) = E[XY] - E[X]E[Y] = 0 - (0)E[Y] = 0$.
- (b) The random variables X and Y are not independent, since $P((X = 0) \cap (Y = 0)) = \frac{1}{5}$ but $P(X = 0)P(Y = 0) = \left(\frac{1}{5}\right)\left(\frac{1}{5}\right) = \frac{1}{25}$.

1. (Adapted from a question in Durrett.) 12 ducks fly overhead. Each of 6 hunters picks one duck at random to aim at and hits it with probability 0.6. Any duck that is hit by a hunter (or more than one hunter) dies. The shots of the different hunters are independent.
- a. What is the probability that hunter 1 shoots at duck 1 and hits duck 1? What is the probability that hunter 1 does not hit duck 1? (In the second question, either the hunter shoots at duck 1 and misses or shoots at another duck.)

The probability that hunter 1 shoots at duck 1 and hits duck 1 is $\frac{1}{12} \cdot 0.6 = 0.05$. The probability that hunter 1 does not hit duck 1 is 0.95.

We can calculate more formally as follows:

Let A be the event that hunter 1 shoots at duck 1, and let B be the event that hunter 1 hits duck 1.

$$P(A \cap B) = P(A)P(B|A) = \left(\frac{1}{12}\right)(0.6).$$

- b. What is the probability that none of the six hunters hits duck 1? What is the probability some hunter hits duck 1?

Let C_i be the event that hunter i does not hit duck 1. Then $P(C_i) = 0.95$. The events C_i are independent. The probability that no hunter hits duck 1 is thus $(0.95)^6$, and the probability that some hunter hits duck 1 is $1 - (0.95)^6$.

- c. Use the method of indicators to calculate the expected number of ducks that are killed.

Let N be the random variable counting the number of ducks that are killed. For $j = 1, \dots, 12$, let D_j be the event that duck j is killed. Let X_j be the indicator for this event. Then $N = X_1 + X_2 + \dots + X_{12}$. We thus have $E[N] = E[X_1] + E[X_2] + \dots + E[X_{12}] = 12(1 - (0.95)^6) \approx 3.1789$.

- d. Use the method of indicators to calculate the expected number of hunters who hit a duck.

For $i = 1, \dots, 6$, let F_i be the event that hunter i hits a duck. Then $P(F_i) = 0.6$. Let Y_i be the indicator of F_i . The number of hunters who hit a duck is then $Y_1 + \dots + Y_6$, and so the expected number of hunters who hit a duck is $E[Y_1] + \dots + E[Y_6] = 6(0.6) = 3.6$. This number is a bit larger than the expected number of ducks killed, since more than one hunter may shoot at any given duck.

